

COLLABORATIVE RESEARCH: HCC: SHF: SMALL: NSF-MUR: DISTRAC-A TOOL FOR TRACKING DISENGAGEMENT IN OPEN SOURCE SOFTWARE PROJECTS

Overview.

Modern IT systems heavily rely on Open Source Software (OSS), which powers everything from critical infrastructure to everyday applications. Yet the long-term viability of OSS projects faces a persistent challenge: contributor disengagement. Many OSS initiatives depend on developers who often shift their focus to career changes, family obligations, or other life priorities. Even when maintained by paid professionals, projects falter when their employers redirect resources or realign business objectives. As a result, numerous valuable open source projects either stagnate with mounting maintenance backlogs or face complete abandonment, risking the broader software ecosystem that depends on them. This proposal addresses the need to model and track contributor disengagement in OSS. In this proposed work, we (1) develop a theoretical model to identify sociotechnical factors related to OSS disengagement and (2) create DISTRAC, an evidence-based tool to predict and assess possible disengagements, fostering the sustainability of OSS projects.

This proposal is a collaboration with the University of Bari under the NSF-MUR MoU, with the following **non-NSF funded collaborators:** Fabio Calefato, Filippo Lanubile, and Nicole Novieli.

Keywords: Open Source Software (OSS), Contributor Retention, Sustainability, Disengagement

Intellectual Merit.

This project addresses the critical challenge of contributor disengagement in OSS projects by developing and evaluating a predictive modeling framework and DISTRAC, a tool designed to assess and simulate the impact of developer disengagement. Grounded in a dynamic sociotechnical framework, DISTRAC integrates a predictive model with contribution patterns, collaboration behaviors, and communication traces to provide maintainers with early warnings and actionable insights for sustaining project development.

Broader Impacts.

This research advances both theoretical understanding and practical support for OSS communities to mitigate disruptions and promote long-term engagement. OSS forms the foundation for many societal activities, and the impact of this proposal may extend well beyond the project's scope. In particular, we have partnered with diverse OSS projects that will help design and evaluate the solutions and will directly benefit from our findings. Moreover, the proposal's impact extends beyond technical contributions. We will ensure broad accessibility through open dissemination of all tools, datasets, and research findings. The project also emphasizes inclusive design principles to promote diversity within OSS communities. Finally, building on our track record of working with underrepresented groups, we will create targeted opportunities for Native American and Hispanic populations. This focus aligns with our lead institution's mission as a Minority Serving Institution and our dedication to fostering a more inclusive technical community.

1 Overview, Objectives, and Significance of the Project

Open Source Software (OSS) forms the backbone of our digital society [28]. It has made technology more accessible globally, empowering companies and developers. However, its decentralized, peer-production nature complicates contributor retention and the securing of financial support for developers [123, 35]. Many organizations use OSS without contributing to its maintenance, jeopardizing its sustainability. Despite these challenges, OSS remains a cornerstone of the modern economy [45]. For example, in 2016, the internet was momentarily disrupted because 11 lines in the OSS JavaScript registry npm were unpublished [61]. This incident is evidence of the reliance of major organizations such as Facebook, Netflix, Airbnb, and the US government [23, 114] on OSS.

A key component of OSS sustainability is fostering a strong and enduring community, which depends on attracting and retaining contributors. Despite significant research into onboarding newcomers and reducing barriers to entry [101, 98, 100], the dynamics of contributor disengagement, particularly among core contributors or maintainers, remain underexplored. The challenges of tracking and managing disengagement are amplified in OSS settings compared to traditional industry environments, due to loosely coupled work patterns, reduced awareness among co-workers, and less centralized hierarchical control and organization [24]. Understanding disengagement is essential to maintaining OSS projects' sustainability [3, 65]. Without effective tools to anticipate and mitigate these risks, maintainers often struggle to manage disruptions caused by unexpected contributor breaks [49]. Addressing these gaps is essential for ensuring the sustainability of OSS projects and fostering resilient development communities.

The problem is especially acute when maintainers or core developers disengage or take extended breaks. When this happens, projects suffer from knowledge silos, stalled progress, and disruptions to workflows, such as code reviews, contributions, and mentorship. High attrition rates among core contributors can lead to stalled project evolution and increase the maintenance burden on remaining developers [5, 3].

Disengagement, in our context, refers to both breaks that developers take from the project (coming back later) and when developers leave the project for good. Disengagement is particularly concerning in critical infrastructure projects, where losing expertise can leave security vulnerabilities unresolved and create risks for dependent systems [13]. The abrupt departure of a maintainer responsible for a widely used module can delay updates, reduce user confidence, and harm the ecosystem's long-term viability. Proactively addressing these challenges requires strategies to understand and assess disengagement, ensuring that projects remain sustainable over time.

This proposal addresses these issues through a multi-thrust approach centered around DISTRAC, a novel **DIS**ENGAGEMENT **TR**ACKER tool designed to assist OSS maintainers with predictive insights and proactive management tools:

- **Thrust 1: Predictive Modeling of Developer Breaks**

This thrust employs predictive models trained on contributor activity patterns to predict when developers are at risk of taking breaks that might anticipate disengagement. The model identifies early warning signs of disengagement by analyzing diverse data sources, such as contribution rhythms, collaboration behaviors, and communication traces, providing actionable insights for maintainers. The model also allows the prediction of the potential impact of specific developers' disengagement.

- **Thrust 2: DISTRAC - A Disengagement Tracker Tool**

Building on predictive insights, this thrust delivers DISTRAC, an interactive tool tailored to OSS maintainers. The tool displays disengagement risks for core contributors and assesses the potential impact of their absence.

- **Thrust 3: Iterative Evaluation in Real-World Settings**

To ensure the effectiveness and adaptability of the proposed tools, this thrust involves iterative evaluation and refinement through field studies with OSS communities. By engaging maintainers in participatory design and testing, DISTRAC will be suitable for diverse project contexts, ranging from large-scale foundations to smaller, community-driven efforts.

As an illustrative example, consider a core developer, Alex, who maintains a critical module of OpenHealthAI, an OSS project used by millions. Alex disengages temporarily (takes a break) without notice, leaving unresolved bugs and unreviewed pull requests. This disrupts the release cycle, frustrates contributors dependent on Alex’s input, and creates a bottleneck in the project’s workflow. Another core developer, unaware of Alex’s break, struggles to redistribute the workload, prioritize tasks, and find people with expertise in the module Alex maintained, ultimately causing delays, increasing the backlog, and diminishing contributor trust.

Using DISTRAC, the maintainer could have received an early warning about Alex’s decreasing activity based on predictive insights. Even if Alex had provided notice, the maintainers—or Alex—could have used DISTRAC to simulate the leave and understand its impact. The community could have taken actions to mitigate risks before Alex disengaged, enabling a proactive response to sustain project stability.

It is important to highlight that DISTRAC focuses on predicting disengagement trends to enhance sustainability rather than targeting individual contributors. The tool is designed as a decision-support system for maintainers to keep the project running in cases of disengagement and enable maintainers to simulate the impact of a disengagement event, promoting inclusive stewardship of open source projects.

This proposal is a transformative step forward in the sustainability of open source software. Aligned with the NSF-MUR solicitation [67], this project unites expertise from the United States and Italy through a collaboration between leading researchers in OSS, machine learning, and statistical modeling in Software Engineering. By combining predictive modeling, participatory design, and real-world evaluations, this proposal delivers actionable solutions for OSS community sustainability.

2 Proposed Work

This project develops a tool based on a predictive model of developer disengagement in OSS to help project maintainers anticipate and understand the impact of developer disengagement.

In the following, we detail the three thrusts planned in the proposal, as presented in Figure 1. In Thrust 1 (Section 2.1), we develop a predictive model that will identify when developers are taking longer than usual breaks and, thus, are at risk of disengaging. Building on this model, in Thrust 2 (Section 2.2), we develop DISTRAC to assess potential project impacts of predicted breaks. Finally, in Thrust 3 (Section 2.3), we conduct summative evaluation through field studies with users applying DISTRAC in real OSS projects.

2.1 Thrust 1: Predictive Modeling of Developer Breaks

In Thrust 1, we will develop and assess a predictive model to anticipate when developers are at risk of taking breaks that anticipate complete disengagement from OSS projects. In a previous work [13, 49], we investigated the inactivity patterns of core developers in OSS projects by analyzing individual contribution rhythms across 18 GitHub organizations. This study revealed that all core developers experience breaks, with 45% disengaging for at least one year. This work relied

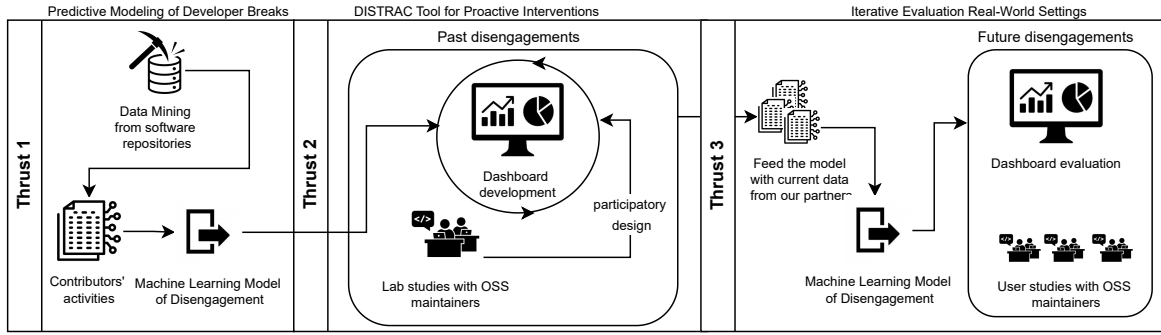


Figure 1: Proposed work overview

on fixed thresholds and binary classifications that did not account for the transitions in developer activity. These limitations highlighted the need for more robust predictive tools and the need to assess the impact on project sustainability.

Therefore, building upon our previous work for detecting breaks based on individual contribution rhythms [13], the predictive model will assess disengagement probabilities and potential consequences. This predictive capability is essential for project sustainability, especially in the case of long-term contributors and TF developers [122, 6], who hold unique knowledge and responsibilities and whose loss would threaten project survival [51, 32, 65]. The predictions generated by our model will drive the development of the DISTRAC tool (detailed in Thrust 2), enabling project maintainers to proactively assess and address potential disruptions before they materialize.

We will focus on studying the breaks and disengagement of Truck Factor (TF) developers, whose departure would endanger project survival [6]. TF is a metric used to assess the ‘resilience’ of a software project by determining the minimum number of contributors whose disengagement would critically impact its survival. As proposed by Avelino et al. [6], it is calculated by analyzing the Degree of Authorship (DOA) for each file, which reflects a developer’s expertise relative to others. The DOA considers factors such as who created the file and the proportion of changes made by each developer. The project’s TF is then determined by identifying the smallest set of developers who are the main authors of at least 50% of the files. A low Truck Factor indicates heavy reliance on a small group of contributors, while a high Truck Factor suggests that responsibilities are more evenly distributed, making the project more robust.

We organize the model development and validation into multiple stages (Figure 2), which we detail in the following subsections.

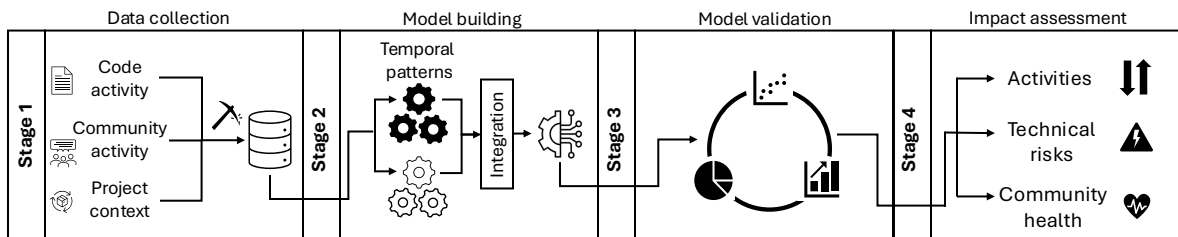


Figure 2: Thrust 1: Model development and validation overview

2.1.1 Stage 1: Mining Activity Traces

Stage 1 establishes the foundation for our break prediction framework by creating a comprehensive dataset of developer activities. This stage encompasses three key steps: (i) establishing a project sample, (ii) collecting a rich set of activity traces, and (iii) contextualizing these activities within project-specific factors.

(i) To ensure the broad applicability of our approach, we will build a comprehensive dataset of OSS projects. In preliminary work [13], we analyzed 18 OSS projects through convenience sampling. For this proposal, we will adopt a stratified random sampling approach to capture diverse contribution patterns beyond GitHub-centric projects. Our sampling framework will consider three key dimensions: core project characteristics (including domain, size, age, and programming languages), community aspects (such as governance model and corporate involvement), and development infrastructure (spanning platforms and communication channels). To ensure data quality, we will follow established repository mining practices [55] by excluding forks, data storage repositories, personal projects, non-software projects, and student assignments, for example. Additionally, we will require projects to have at least two years of activity and two Truck Factor developers, thus ensuring the analysis of established projects with sufficient historical data, for which contributor disengagement poses significant organizational risks. We will include abandoned projects in our sample, as these could provide valuable historical insights into break patterns and disengagement factors that may have led to project discontinuation. This sampling approach will yield a larger yet manageable dataset that captures diverse development environments and community structures.

Besides searching public datasets [37, 11], we will also collect data from our partner organizations (Open Source Initiative/ClearlyDefined, JabRef) (see letters) and will reach out to other organizations we have connections (e.g., Apache Software Foundation [29], Linux Foundation [46, 108, 111, 110], and GitHub [60, 59, 22]). These projects and organizations are important due to the involvement of their members with our proposal. They will provide feedback on the model's outcomes and ensure we collect data from the appropriate sources.

(ii) We will collect data that enables us to go beyond the basic metrics captured in our previous work [13], which analyzed individual contribution rhythms through coding activities (commits, pull requests) and non-coding activities (issues and pull request discussions) in GitHub. We will introduce new categories of metrics that can serve as early warning indicators of potential breaks. This expanded set of metrics will provide a more comprehensive view of developer engagement patterns and potential early disengagement signals [62, 65]. The first category focuses on code-related metrics and incorporates detailed complexity metrics such as the number of files and lines modified per commit, code complexity metrics [74], and evolving code ownership patterns [31, 33]. The second category includes communication metrics across diverse channels, tracking response times, interaction frequencies, and sentiment changes that might indicate shifting engagement levels [58]. The third category focuses on sociotechnical network analysis metrics [120, 54, 53] to capture developers' roles and relationships within their communities. This set of metrics will help us construct comprehensive interaction graphs encompassing both formal project activities (e.g., code reviews and issue discussions) and more informal communications across various platforms like GitHub, GitLab, mailing lists, and public chat servers like Discord.

(iii) Finally, we will contextualize these individual activity traces within broader project ecosystem factors that may influence break patterns. This includes analyzing project dependency networks [25] to understand maintenance complexity, examining project maturity indicators such as size and age [66], and considering organizational factors like corporate sponsorship [123, 12]. We will also evaluate community health metrics from the literature [52, 38]. These contextual factors

will help enhance our break prediction model by accounting for the project-specific environments and constraints that shape contributor behavior and engagement patterns.

2.1.2 Stage 2: Building the Prediction Model

The second stage focuses on developing a predictive model to anticipate TF developer breaks using the activity traces collected in the previous stage. Our proposed approach combines multiple analysis techniques to create a robust prediction framework that addresses the potential limitations of our previous far-out values approach [13]. Specifically, the previous approach has three key drawbacks: (i) it provides binary break classifications rather than nuanced risk assessments (e.g., labeling a developer either as taking a break or not, without early warnings or risk levels); (ii) it uses fixed thresholds that do not account for natural variations in development activity due to seasonality and project cycles (e.g., treating a two-week gap close to a milestone release the same as a two-week gap during a holiday season); and (iii) it only considers temporal gaps between activities rather than changes in contribution behavior (e.g., missing when developers gradually shift from writing code to only reviewing others' work before disengaging). These limitations prevented our previous approach from capturing the varied range of warning signs that may anticipate upcoming disengagement.

To build the enhanced break prediction model, we will implement two complementary strategies: building customized models for individual projects and developing a generalized cross-project solution. To build the customized models, we will first establish project-specific baseline activity patterns for each TF developer through a combination of complementary methods. For the generalized solution, we will develop a general pre-trained model that can work across projects, enabling quick risk assessment even for projects with limited historical data or those wanting to avoid data collection and model training overhead.

Given that our previous work [13] demonstrated the highly individualized nature of contribution rhythms, we will implement an ensemble of time series modeling techniques to capture patterns of developer behavior. Traditional time series approaches like ARIMA [90] excel at modeling regular, linear patterns and seasonal variations — for example, they can help us capture regular patterns of developers consistently making more commits on Tuesdays and Thursdays across multiple weeks or months. Exponential smoothing methods [34] assigns differential weights to temporal data points, with greater emphasis on recent observations and diminishing influence on historical data. These methods can be particularly effective in time series analysis of developer contributions — for example, they could help us detect recent patterns like a sustained decrease in weekly commit frequency while naturally reducing the impact of older variations such as historical spikes in activity from past release cycles or one-off contribution bursts. Other techniques, such as LSTM, can capture sophisticated non-linear patterns and long-term dependencies across multiple activity types [91] — for example, LSTM networks can help us identify complex patterns where developers first show an intense period of commits while implementing new features, then transition to primarily fixing bugs, and finally increase their involvement in pull request discussions.

Finally, an integration layer forms the foundation of our approach for building a predictive model of developer breaks. This layer first combines predictions through weighted averaging [124], integrating signals from multiple models. The ensemble weights will adjust dynamically based on context, emphasizing models that excel at predicting irregular activity during seasonal periods like holidays, while favoring those better suited for capturing sudden contribution bursts during project milestones. Building on the synthesized predictions generated by the ensemble, the integration layer will then produce 'break risk scores' with confidence intervals, estimated timeframes for developers' potential breaks, and key contributing factors.

This multi-step approach integrates different analytical techniques that excel at complementary aspects of prediction — from identifying regular contribution rhythms and adapting to recent trends, to capturing complex long-term patterns of interaction — thus accounting for natural variations in contribution patterns across different timescales, from weekly cycles to seasonal trends. Overall, we argue that this comprehensive approach improves upon binary break classification by providing contextual predictions that account for both project-specific factors and developers’ individual contribution patterns.

2.1.3 Stage 3: Model Validation

Stage 3 involves two complementary phases of statistical analysis to validate and refine our prediction model’s technical performance and theoretical soundness before implementing DISTRAC in Thrust 2 and field testing in Thrust 3.

The first phase centers on statistical validation using the data collected during our mining stage. Following standard machine learning practices, we will partition the collected data into training, validation, and test sets to ensure robust evaluation of our model’s predictions. We will evaluate our model by comparing its predicted risk levels and timing of breaks against actual developer break patterns, measuring how accurately it anticipates both the likelihood and the time window. Additionally, we will validate that our model performs consistently across different developer groups and project contexts. LSTM-based RNNs currently do not verify predicted values against actual values during the training and testing phases of the model. Implementing a verification step can enhance the accuracy and reliability of the model, leading to better results. Generative Adversarial Networks (GAN) combined with Convolutional Neural Networks (CNN) may work as generative and discriminative models, respectively, to verify LSTM predictions [56].

In the second phase, we will develop a series of regression models to examine how different factors contribute to disengagement likelihood. These analyses will incorporate directly measurable variables (such as commit frequency and issue resolution times) and composite activity metrics, which we will construct through dimensionality reduction techniques (such as Principal Component Analysis [1] or Non-negative Matrix Factorization [57]). These techniques will help us identify meaningful patterns in developer behavior (e.g., ‘review engagement’ combining review frequency, length, and response time, or ‘code contribution intensity’ incorporating commit size, frequency, and complexity) while reducing dimensionality and avoiding multicollinearity.

Since repository metrics typically follow heavy-tailed distributions [9], we will employ appropriate statistical techniques to handle non-normal distributions and outliers in our data. This may include variable transformations or robust regression methods, chosen based on the specific characteristics of our dataset and the assumptions that best fit our prediction context.

The validation framework will integrate results from these regression analyses in three steps. First, we will establish baseline models using only direct activity metrics. Second, we will incorporate the composite metrics to capture higher-level activity patterns. Finally, we will add project-level characteristics and their interactions. At each step, we will assess model improvement through standard goodness-of-fit metrics on the validation set, reserving the final held-out test data for evaluating only the best-performing model configuration. This systematic approach will help us understand both the individual and collective contribution of different factors to break prediction while maintaining the interpretability of results.

Finally, throughout the validation process, we will pay particular attention to how model performance varies across different types of contributors, including paid versus volunteer developers, contributors from different geographic regions, and those in various roles within the sampled projects. This stratified analysis will help ensure that our model provides reliable predictions for diverse contributor backgrounds and project contexts.

2.1.4 Stage 4: Impact Assessment

Building on the break predictions generated in *Stage 2* and validated in *Stage 3*, this stage focuses on assessing the potential consequences of predicted breaks on project sustainability. The impact assessment stage analyzes three key dimensions affecting project health when contributors take longer than usual breaks or disengage.

The first dimension examines activity drops through statistical analysis of historical break patterns and their effects. We will develop regression models to quantify the relationship between types of breaks and various contribution metrics, accounting for contributor characteristics, break attributes, and project-level properties. These models will help assess both direct effects (reduction in the breaking contributor's activity) and indirect effects (increased workload on remaining team members).

The second dimension focuses on knowledge distribution and technical risk. First, we will identify critical components that may become under-maintained during breaks using code metrics (e.g., complexity [74], ownership [33], and churn [44]) as well as dependency graphs [125]. Next, we will analyze the contributors' history and build their profiles to assess the human resource impact when disengagement occurs. This will help identify potential risks for project stability.

The third dimension examines community health impacts through sociotechnical network analysis. By modeling how breaks affect collaboration patterns, mentorship relationships, and information flow within the project, we can estimate the social capital at risk and identify where additional community support may be needed. This analysis will be particularly valuable for projects with high proportions of volunteer contributors (such as non-sponsored projects) or those heavily dependent on specific maintainers for community management (such as projects with a low TF number). Finally, we will evaluate the workforce needs by summarizing the predicted loss in the talent pool to anticipate future recruitment process needs (as detailed in Thrust 2).

To validate the impact assessment models, we will analyze historical cases of contributor disengagement in our sample projects, comparing our predicted impacts against observed effects on project metrics and community health indicators. This process will help calibrate the models and ensure they provide relevant information for project maintainers. The resulting impact predictions will directly inform the design of DISTRAC in Thrust 2, helping maintainers prioritize preventive actions based on both the likelihood and potential consequences of predicted breaks.

2.1.5 Deliverables

The primary deliverable of Thrust 1 is a comprehensive break prediction framework combining time series modeling and impact assessment capabilities. To help OSS projects analyze contributor activity patterns and predict disengagement, all the source code employed to build the model and the model parameters will be released as open source. All the analytical components will be modular, allowing communities to adapt and extend them based on their specific needs and data sources. The general pre-trained model will be uploaded to Hugging Face or a similar platform for easy access. We will release an open dataset enabling researchers to conduct further analyses and studies in related domains.

We will publish at least two empirical studies: (i) one that analyzes the effectiveness of different modeling techniques in predicting breaks across various project contexts and contributor demographics and (ii) another that examines the relationship between predicted disengagement and project health metrics, providing actionable insights for community managers. These research outputs will establish a scientific foundation for developing our disengagement tracker in Thrust 2.

2.1.6 Risks

The main risk in Thrust 1 is that the predictive model may fail to generalize across diverse OSS projects due to variability in activity patterns and data quality. Additionally, false positives could erode trust in the tool, while false negatives may lead to missed interventions. In this sense, we will ensure a diverse sampling of projects to understand to what extent it is possible to generalize a single model. More importantly, the goal here is to provide a framework that enables the creation of project/ecosystem-specific models, aligning with their data sources and development rhythm. We will validate the model iteratively with real-world data and incorporate feedback from maintainers to refine predictions and provide guidance on how to customize for other projects.

2.2 Thrust 2: Developing DISTRAC – a Disengagement Tracker Tool

In Thrust 2, we develop an interactive tool called DISTRAC (**DIS**ENGAGEMENT **T**RACKER) to aid OSS project maintainers by providing real-time predictive insights into potential contributor disengagement. Building on the prediction model established in Thrust 1, DISTRAC will empower maintainers to proactively manage community dynamics and continuity by anticipating the likelihood and impact of contributor leaves. DISTRAC’s key features include predictive displays, impact assessment, simulation capabilities, and dependency mapping, each designed to address maintainer needs for efficient community management and resilience.

The tracker will highlight contributors’ likelihood of disengaging (taking a break or leaving), enabling maintainers to see how the project may be at risk. Alongside this, we will provide an impact assessment functionality to simulate the consequences of a contributor’s absence, according to metrics such as committed Lines of Code (LoC), pull request activity, issue interactions, module or file ownership, and the contributor’s expertise with APIs [88]. This analysis will be instrumental in identifying critical dependencies and assessing the potential impact on the project’s stability and progress. The simulation tool will allow maintainers to manually input hypothetical disengagement for specific contributors, enabling them to understand the potential impact of such scenarios. This feature is useful, for example, when contributors informally communicate planned absences, allowing maintainers to prepare for possible disruptions.

2.2.1 Participatory Design Process and prototyping

We will engage in participatory design sessions with OSS maintainers to ensure DISTRAC aligns with real-world needs. These sessions will provide a collaborative platform for maintainers to share their experiences, preferences, and expectations for DISTRAC. During these sessions, we will present them with established tools and strategies from existing literature (e.g., Git Truck [48], Maintainers Dashboard [43], Climate coach [81] and CommunityTapestry [41]), helping them identify desirable features and functionalities specific to OSS project management. The maintainers will actively collaborate in designing the tool’s navigation and usability, ensuring it meets their practical requirements for ease of use and effective data presentation.

Following the initial design phase, we will proceed with iterative prototyping and formative evaluations, incorporating maintainer feedback at each stage. This process will involve developing successive prototypes, from wireframes to functional designs, and gathering insights from maintainers to refine usability, functionality, and visualization. During each evaluation iteration, we will integrate a set of evaluation methods, including usability testing [68], cognitive walkthroughs [30], and scenario-based assessments [39] to explore how maintainers interact with different features. Each evaluation cycle will involve feedback collection sessions, ensuring that feedback is systematically incorporated to enhance the tool’s usability and effectiveness. We aim to

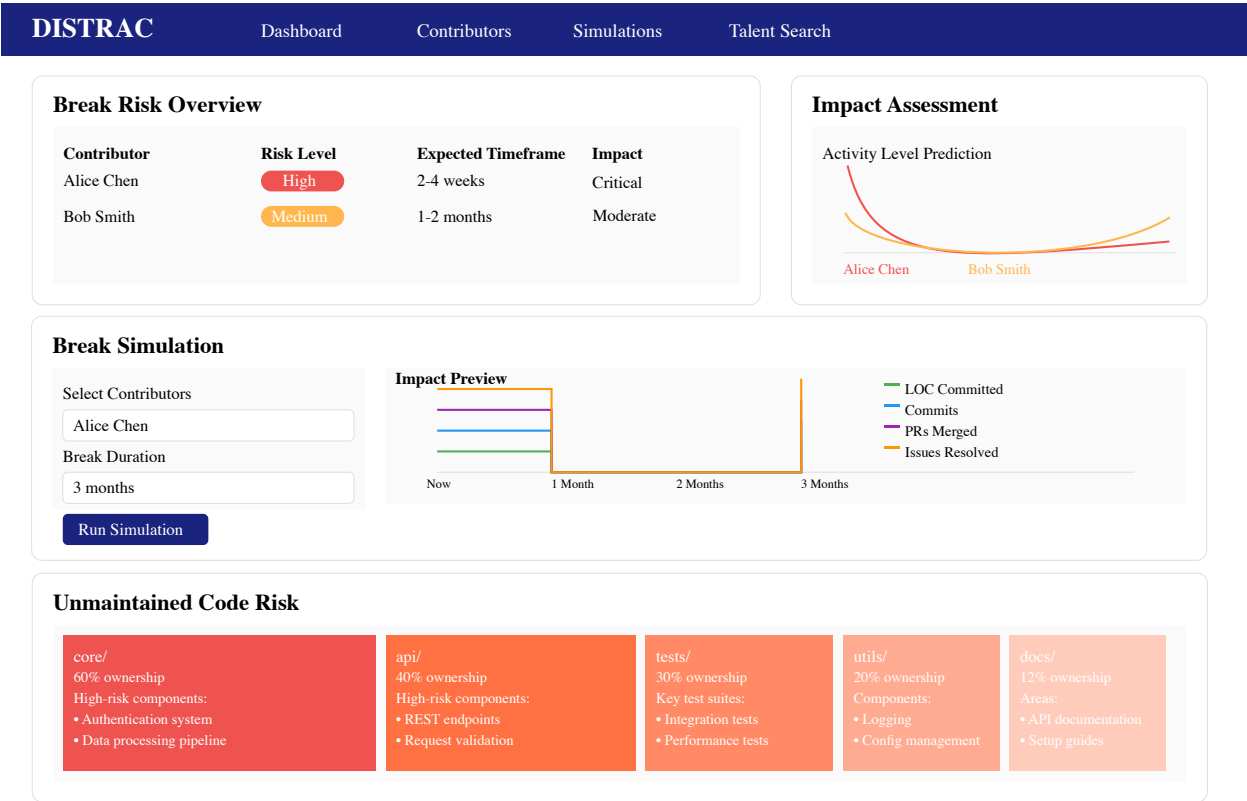


Figure 3: DISTRAC mockup

complete at least three iterative cycles, adjusting and refining the prototype based on feedback to converge on a functional, useful, and intuitive design.

A mockup of DISTRAC is shown in Figure 3. In our vision, the tool will show the disengagement risk and the expected timeframes for TF developers (top left), predicted activity trends (top right), simulated impacts of potential breaks (middle), and code ownership visualization showing modules at risk of becoming unmaintained (bottom).

2.2.2 Lab Studies

After the participatory design prototyping, we will develop a robust version of the tool and conduct lab studies with maintainers, focusing on retrospective analyses of historical project cases. As part of the lab studies, maintainers from partner projects (Open Source Initiative/ClearlyDefined, JabRef) will review specific past disengagement events within their OSS projects, such as contributor breaks or sudden departures, examining these events through the lens of the tool’s risk and severity predictions using the data that was available at that time. This reflective approach will enable maintainers to evaluate the tool’s accuracy in understanding high-risk scenarios, considering both the probability of the predicted break and the anticipated impact.

Maintainers will be instructed to reflect on the actual outcomes of these events, analyzing where the tool’s indicators matched or diverged from reality. Key questions will guide the maintainers’ reflections: Was the risk accurately flagged? Were some significant risks underestimated or overlooked? Was the impact adequately assessed in terms of activity drops, module maintenance, or community interaction? Did the interface help understand the risks and impacts of the

disengagement? By exploring these questions, maintainers will provide insights into the tool's predictive validity and areas for improvement. We will work iteratively on improving the tool as we find mismatches in perceptions and what the tool reports. New studies will be conducted to evaluate the refined version of the tool.

In these lab studies, we will also probe the impact of DISTRAC by inviting maintainers to consider hypothetical actions they might have taken had they been aware of a potential disengagement in advance. For instance, maintainers will be asked to reflect on whether they could have redistributed tasks, replaced contributors, or prepared strategies to mitigate disruption. This exercise seeks to reveal whether the tool could influence project management and sustainability and provide additional feedback on its design. Such reflections are important to understanding how predictive data can enhance proactive decision-making and where DISTRAC might provide additional context or information.

2.2.3 Deliverables

The primary deliverable of Thrust 2 is DISTRAC, offering maintainers a data-driven tool for predicting and simulating potential disruptions due to contributor breaks. In addition to DISTRAC, we will produce at least two scholarly outputs. First, we will document the tool development process and participatory design findings, contributing to software engineering literature. We will also work on an empirical paper reporting the maintainers' perception of the information displayed. This second paper will be based on DISTRAC evaluation in historical settings, capturing maintainers' qualitative feedback.

2.2.4 Risks

For Thrust 2, the main risk lies in the usability and utility of DISTRAC. If the tracker is not intuitive or recommendations lack contextual relevance, maintainers may avoid using it. We will address this by conducting participatory design sessions with maintainers from our collaborators and using iterative usability testing to ensure the tool meets their needs. Moreover, we will monitor performance bottlenecks or technical flaws that could prevent the tool from being fully functional. We will address this by building prototypes iteratively, starting with basic functionalities and gradually adding complexity, and also conducting regular testing and optimization throughout development.

Finally, there is a risk of not being able to recruit maintainers to participate in the tool design. We will address this by actively partnering with multiple OSS projects. Besides our partners, we will also propose talks to present the tool during Open Source conferences, such as Open Source Summit and Linux Plumbers, where we had already given talks, to recruit more participants.

2.3 Thrust 3: Evaluating the impact of the approach in practice

In Thrust 3, we focus on summative evaluation by deploying DISTRAC in real-world OSS projects to assess its effectiveness and maintainers' experiences. Partnering with OSS projects, we will integrate DISTRAC into active projects and collect data across multiple perspectives to triangulate findings.

2.3.1 Summative Studies

First, we will conduct end-user field studies with core members and maintainers of our partner OSS projects to evaluate DISTRAC's performance and usability in real-world scenarios. These

studies will be followed by debriefing interviews to gather detailed feedback on their experiences using DISTRAC. This step focuses on assessing DISTRAC’s effectiveness in its deployed state, examining its ability to provide meaningful predictions for mitigating disengagement impacts. By observing how participants use the tool and eliciting their perceptions of its utility, we aim to determine whether DISTRAC meets its intended objectives in practice. Specific areas of evaluation include the perceived relevance of the predictions and the overall impact on maintainers’ decision-making processes. To ensure rigor, we will analyze this data using qualitative methods with inter-rater reliability checks, triangulation, and member checking, which allow us to validate our findings and draw reliable conclusions about the tool’s impact.

Second, we will ask maintainers and core members to assess how DISTRAC helped prevent disengagement or mitigate its consequences for their projects. The goal here is to measure the outcomes and impacts of DISTRAC *in-vivo*, analyzing how the outputs of Thrust 1 and Thrust 2 work in real scenarios. Periodic surveys will gather evidence on whether maintainers found DISTRAC useful for managing contributor turnover and whether its recommendations were actionable and effective. These surveys will also identify areas for improvement and document cases where DISTRAC was less successful. The summative approach ensures we evaluate DISTRAC based on real-world results, drawing insights into its overall contribution to sustainability.

Third, we will conduct a 1-year longitudinal study with core developers and maintainers who adopt DISTRAC for their projects. This summative step evaluates the sustained effectiveness of DISTRAC over time by collecting usage metrics (see Table 1) at regular intervals. By analyzing long-term trends in tool adoption, feature usage, and community outcomes, we assess DISTRAC’s durability as a solution for contributor disengagement. The focus here is to understand how DISTRAC performs under extended use, seeking evidence of the prediction model and the GUI in fostering project continuity and reducing disengagement risks and impact.

To analyze the collected data, we will employ Regression Discontinuity Design (RDD) [104, 50], which has been successfully used in software engineering contexts [121, 19]. RDD will enable us to model the discontinuities in project outcomes at the point of intervention (i.e., the adoption of DISTRAC) and evaluate their persistence over time. This method provides robust evidence of the causal impact of DISTRAC on key OSS metrics. Our team’s expertise in mining software repositories and applying RDD ensures reliable and actionable insights [119, 118].

Recruitment. We will recruit maintainers and contributors from our partner projects and additional OSS projects outside our immediate collaborators. Involving external projects ensures diverse perspectives and increases the generalizability of our findings. We aim to include 6–10 external projects per evaluation cycle, encompassing various sizes, domains, and contributor demographics. Recruitment efforts will prioritize diversity in participants’ tenure, expertise, and backgrounds, ensuring representation of traditionally underrepresented groups. Besides extensive advertisements, we will give talks at open source conferences, as we have done in the past.

Measurements. We will use validated metrics and instruments to the extent possible, as detailed in Table 1, encompassing engagement, satisfaction, usability, effectiveness, and demographic characteristics.

2.3.2 Deliverables

The primary deliverable for Thrust 3 is a comprehensive understanding of DISTRAC’s impact on real OSS projects, focusing on its effectiveness in predicting and mitigating disengagement, supporting maintainers’ decision-making, and promoting community sustainability. This final evaluation phase will provide robust empirical evidence of DISTRAC’s utility through field studies and longitudinal analysis.

Table 1: Metrics and instruments for field user evaluation.

Construct	Metrics, instruments, procedures, and literature
Temporal analysis of activity indicators	RDD [104], & health, sustainability, quality metrics [20]
Usage of the technology	perceived usefulness, perceived ease of use, behavioral intention, frequency of usage, time spans, and # of developers [116]
Individual characteristics	Computer self-efficacy, computer anxiety, computer playfulness, perceptions of external control [115]
System characteristics—related adjustments—	Perceived enjoyment, objective usability [115]
Maintainers perceptions	interviews
Engagement	User Engagement Scale (UES) [72]
Demographics	Validated instruments [4, 71, 10]

At the end of this thrust, we will have an improved version of DISTRAC, informed by the findings of field studies and longitudinal evaluations. This version of DISTRAC will incorporate adjustments and refinements based on real-world feedback to ensure its applicability and reliability across diverse OSS contexts.

The findings from this thrust will culminate in at least two peer-reviewed publications. One will document the results of the field studies, highlighting maintainers’ experiences, their perceptions of DISTRAC, and their role in addressing disengagement challenges. The other will present insights from the longitudinal evaluation, incorporating both quantitative metrics and qualitative feedback to assess the sustained impact of DISTRAC on project outcomes. These publications will provide critical contributions to the literature on OSS sustainability and maintainer support.

2.3.3 Risks

In Thrust 3, the primary risk is the challenge of recruiting enough OSS projects for meaningful evaluation, especially those experiencing disengagement events. To mitigate this, we will leverage our partnerships and community connections to recruit diverse projects and incorporate simulated disengagement scenarios into the evaluation process. This dual approach ensures comprehensive testing and feedback even if real-world disengagement events are limited. Alternatively, we will run more lab studies if necessary.

Another potential risk is that DISTRAC may be deemed not very helpful by maintainers or fail to integrate seamlessly into their workflows. If this occurs, we will conduct in-depth follow-up interviews to understand their concerns and reassess the tool’s value proposition. This result will be relevant for subsequent advances in the field.

3 Team Qualifications

The proposed project is well-suited to our team’s combined expertise and collaborative history. Our team combines expertise in open source software research, statistical modeling, and machine learning, ensuring a robust approach to addressing the challenges of OSS contributor disengagement. A project plan is provided in our Collaboration Plan (see supplementary document).

PI Steinmacher is a leading expert in onboarding, retention, and disengagement in OSS. His research spans critical areas, including newcomer integration [96, 97, 102, 95, 99, 79, 101], disengagement [13, 49], and the role of maintainers in OSS [26]. He has also explored the educational applications of OSS [78, 77, 84] and has extensive experience mining software repositories, applying statistical modeling and machine learning techniques in the OSS domain [118, 119, 109, 110, 85, 87], and conducting CSCW research [107, 8, 98, 117].

PI Trinkenreich consults for Linux Foundation Research [46, 108], and specializes in studying OSS communities and teams dynamics, focusing on OSS contributors' careers [107], motivations [35], sense of belonging [110], and future goals [106]. Her extensive expertise in statistics and predictive models [112, 109, 105] and awarded research brings valuable insights into the human factors driving OSS engagement and retention, providing a complementary perspective on addressing disengagement and fostering sustainable communities. As an active member of The Linux Foundation Research, her network of contacts in OSS will be pivotal in recruiting participants for the proposal.

Co-PI Santos brings additional expertise in machine learning for OSS, ensuring the project's predictive modeling and statistical analysis are grounded in state-of-the-art methodologies. His technical proficiency will contribute to developing and refining DISTRAC and validating its effectiveness across diverse OSS contexts. His studies include strategies to onboard newcomers [86], predicting skills to solve tasks [87], and using social network analysis to support newcomers onboarding [85].

The research labs at the two US institutions provide complementary strengths in OSS research, human factors in software engineering, and diversity and inclusion, creating a well-rounded team. Additionally, this project is a collaborative effort between U.S.-based researchers and a team from the University of Bari in Italy, established under the NSF-MUR solicitation [67]. The NSF-MUR initiative promotes US-Italy collaboration by having MUR funding Italian researchers via a *"single collaborative proposal that will undergo a single review process at NSF"*.

The Italian team, led by Prof. Fabio **Calefato** as **PI** in partnership with Prof. Filippo Lanubile and Prof. Nicole Novielli, contributes expertise in statistical modeling, machine learning, and OSS research. Their involvement ensures a robust and interdisciplinary approach to addressing the challenges of OSS contributor disengagement. The University of Bari team has a strong track record of working on advanced analytical techniques, including predictive modeling and regression analysis [14, 15, 16, 70], making them valuable partners in realizing the goals of this project. This collaboration exemplifies the mission of the NSF-MUR program [67] by combining the complementary strengths of the U.S. and Italian teams to advance research in OSS sustainability. The Italian team has also translated their academic expertise into practice through the establishment of PeoplewareAI [76], an academic spin-off specialized in collaborative software engineering, SE4AI, and machine learning, directly supporting the project's goals by developing people-centered AI solutions to improve OSS sustainability and contributor engagement. The Italian research group also has a strong track record of securing funding for projects through MUR, including EmoQuest (Prin 2014), EMPATHY (Prin 2017), QualAI (Prin 2022), and Digital Lifelong Prevention (PNC-PNRR 2022), a large project with an overall budget of €9M. All these accomplishments demonstrate the team's readiness and suitability for the NSF initiative by showcasing their ability to tackle complex research challenges, innovate in AI [69, 83] and OSS fields [13, 18], and deliver impactful solutions [17, 82].

The US team will lead the development of DISTRAC, including its design, implementation, and validation through controlled studies. They will also oversee the recruitment of participants from OSS communities and the dissemination of results through publications and workshops. The Italian team will mainly focus on advanced predictive and statistical modeling, contributing

to the predictive framework underlying DISTRAC. They will also coordinate with European OSS projects for additional case studies and data collection.

To address the risk of a counterpart not fulfilling their responsibilities, we will have the team working in collaboration during the whole project, with one institution leading each phase but, in co-responsibility with their counterpart in the other country. We have included detailed milestones and deliverables in the Collaboration Plan (see Supplemental Document). Regular bi-weekly meetings will ensure progress is monitored and any issues are promptly addressed. If necessary, the US team has sufficient expertise in predictive and statistical modeling to absorb these tasks, ensuring the project remains on track regardless of unforeseen circumstances.

The research team has a proven history of successful collaboration. The PIs have co-authored multiple high-impact publications, including studies on motivations in OSS (ICSE 2021 [35]), gender biases in open source (TSE 2020 [73], ICSE 2023 [110], ICSE 2024 [105], ICSE 2025 [111]). Dr. Steinmacher and the University of Bari team have previously collaborated on OSS disengagement research, producing a foundational study that inspired this proposal [13, 49]. While the prior work identified disengagement patterns through retrospective analysis, we will provide comprehensive risk assessments instead of binary classification (break/no break), move to dynamic thresholds since fixed failed to account for seasonal or project-specific variations, and focus on gradual shifts in contribution behavior. Moreover, the current proposal goes further by developing a predictive framework and actionable tools for maintainers.

This research is also supported by OSS projects and organizations, namely Open Source Initiative/ClearlyDefined, JabRef(see letters from our collaborators) and the PIs keep collaborating with other organizations that are central in the OSS ecosystem (e.g., Apache Software Foundation [29], Linux Foundation [46, 108, 111, 110], GitHub [60, 59, 22]).

4 Intellectual Merit

This project advances the understanding of contributor disengagement in OSS projects by systematically investigating it as a sociotechnical phenomenon and by gaining knowledge in developing a tool. The research combines empirical studies with validated instruments to model the sociotechnical factors driving disengagement, offering actionable insights for maintainers. Additionally, by leveraging machine learning and participatory design, the project will create DISTRAC, a tool for predicting and mitigating contributor breaks.

The predictive framework and disengagement tracker will provide a transformative contribution to the field, equipping OSS maintainers with tools to address sustainability challenges proactively. Our proposal integrates predictive analytics, impact assessment, and real-world assessment, ensuring the results are both solid and practical. Additionally, the project addresses gaps in existing research by focusing on long-term contributors and maintainers, whose disengagement poses significant risks to OSS projects.

5 Broader Impacts

The project's outcomes will benefit society by promoting the sustainability of OSS. By equipping maintainers with tools to predict and assess contributor disengagement, this work will strengthen OSS projects' stability and continuity, reducing the risk of disruptions to critical software systems. Such improvements will result in improvement throughout the entire tech sector. OSS systems are an important economic driving force [89, 36], benefiting several sectors of society [27, 2, 75], including the scientific community [80].

The proposed DISTRAC tool will empower OSS maintainers with evidence-based approaches to predict and mitigate contributor disengagement, reducing disruptions and maintaining momentum in OSS development. The open-source release of DISTRAC and datasets will not only benefit maintainers but also enable researchers, educators, and practitioners to build upon the project’s findings, fostering collaboration and innovation across the OSS ecosystem.

Besides the direct effect on important OSS communities (see letters), we propose extensive outreach efforts for other researchers, educators, OSS community members, and students who may be interested in our tooling or conceptual contributions. We will involve these populations in the research through seminars, and all the produced code and material will be open-sourced or in Creative Commons. We will encourage and support the reuse of our model, infrastructure, software components, prototypes, and other contributions by applying previously established guidelines [101].

Finally, our work has a diversity and inclusion component by facilitating diversity in communities, advancing diversity in research, and preferably recruiting students from minorities. Northern Arizona University is designated as a Minority Serving Institution (Hispanic serving and American Indian serving) [94, 113], and we will reach these students via institutional channels while recruiting for research and mentoring opportunities.

Dissemination: We will disseminate the results of our work to multiple audiences. For *academics*, we will submit scientific papers and demos to top-tier conferences in the software engineering and OSS fields. For a *large audience*, which might include academics, educators, OSS developers, and students, we will: (i) create a project-specific website to include all publications, software, and materials; and (ii) a Fosstodon (FOSS Mastodon) account [63] to announce findings, events, and recruitment opportunities. (iii) We will also promote synchronous and asynchronous online seminars to discuss the project and how to extend the results to other contexts.

As a measurement of success, we expect to produce at least 4 scientific papers during the process and a final paper after the summative efforts, 2 talks at practitioner/community-oriented gatherings, 3 blog posts, 1 website, 1 open-source project, 3 videos, 1 Fosstodon post per month, and 5 seminars.

6 Results from Prior NSF Support

PI Steinmacher concluded a grant entitled “Gender-Inclusive Open Source through Gender-Inclusive Tools” (IIS-1900903 \$491,098, 08/01/2019 - 07/31/2023). **Intellectual Merit:** This grant applies GenderMag to OSS infrastructure and develops best practices for fixing inclusivity issues. **Broader Impacts:** More equitable contribution opportunities by removing inclusivity bugs. **Relation to this proposal:** The proposal’s topics are closely related to OSS sustainability, focusing on different dimensions. **Publications and Availability:** Several publications have been produced under this award [7, 26, 93, 35, 92, 107, 21, 64, 103, 42, 47, 73, 118, 7, 110, 105, 40] receiving best paper awards on ICSE-SEIP 2022, ICSE 2023, ICSME 2020, and OpenSym 2020.

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Data Management Plan

The PIs—Igor Steinmacher and Bianca Trinkenreich—will take shared responsibility and provide oversight for the management of data associated with the proposed research. All researchers (graduate students, any undergraduate and high school students who may join) associated with this project will be briefed on and required to adhere to the data management plan. In addition, all researchers associated with this project will receive IRB training and certification that includes information and best practices related to the ethical management of human subjects data. The PI will work under the auspices of an IRB protocol.

The international collaborations from the University of Bari will have to follow their regulations and the US institutions' IRB guidance. They will seek approval from their local institutions. Human subject data collected in the US institution will not be shared with the Italian collaborators until a collaborative protocol is approved, and vice-versa.

Data Generated and Access. Data to be gathered or produced in this project will include data from user studies, including field notes, digital audio and video recordings from interviews and lab studies with human subjects, digital audio and video recordings of human subjects, screen captures of study sessions, transcriptions of interview results, and survey data. Additional meta-data will include coding schemes and memos, experiment protocols, design recommendations, derivative design ideas, and scenarios of our proposed tool. The project will also include data collected from repositories. This data will be suitably anonymized, such that participant location or identity is never recorded. Human subjects' data that include privileged and confidential information and raw data will be kept secure in accordance with approved IRB protocols and best practices for the protection of human subjects. Any human subject data that is publicly released or shared across universities will be suitably anonymized and aggregated to maintain confidentiality and protect the privacy of subjects and industry partners.

Dissemination and Sharing of Results. Key excerpts of anonymized primary data will be routinely shared in the context of publications of this research. Final coding schemes, design recommendations, and design prototypes that serve to synthesize across and characterize our holistic understanding of the primary data will be shared as well, in the context of publications of this work. All publications resulting from this research will include relevant methodological information necessary to understand how the data was collected and analyzed. In the event that key data has not been published in formal venues within five years of the termination of this grant, all unpublished manuscripts will be made publicly available as tech notes or working papers.

We will promptly prepare and submit for publication, with authorship that accurately reflects the contributions of those involved, all significant findings from work performed in this project. To promote the widespread use and dissemination of our research results, our first choices for publishing this research will be at venues that allow authors to make digital preprint copies of their publications freely available via their websites. To the extent that this is possible, then, all publications (including tech notes and working papers) resulting from this research will be made publicly available on the PIs website as PDF files.

We will create a project website that will serve as a clearinghouse of our project results, software, and curriculum results. This website will host our final and intermediate results (papers published in conferences as well as technical reports), and links to the available source code (and hosting site) and datasets. The website will also host curriculum materials that will be shared by the team and made available to other educators. We will also add to this site replication package to improve the verifiability, reproducibility, and replicability of our methods and results.

Software. Software prototypes created through this proposal will be made open source under the appropriate Open Source (OSI-approved) license. We will host our project on GitHub, which

will allow our team to coordinate our development efforts, as well as allow the general public to download the prototype as they see fit.

Data Archival. Project results will be archived at each university on their local servers, managed and supported by their department IT team. Individual human subjects' data will be archived on a password-protected system or in locked cabinets accessible to only the PI and their research team. Raw data will be archived for a limited period of no more than three years from the completion of the project as required by the human subjects protocol; after this period has expired, the data will be destroyed to protect the privacy of the individuals involved.

Mentoring Plan

The project will include 2 graduate students. The professional development of graduate students will be supported in different ways. The ultimate goal is to ensure that students acquire technical, scientific, and professional skills to succeed as researchers. Before diving into details, we recognize that no “one size fits all” approach to mentoring will work for all students. It is important to consider the individual differences, needs, backgrounds, and circumstances that change mentoring. The expected mentoring activities for graduate students in this proposal are as follows:

Individual orientation. We believe that the mentor-mentee relationship must be close. From the beginning, we will make clear that the PIs follow an “open door” approach, in which the students are welcome to stop by at their offices when necessary. That said, we will have frequent meetings (2-3 times a week) in the first month of joining. The topics will include the expectations, defining the timeline, topics of study, research methods, interaction with the lab members, processes, and ethics in research. After that, we will keep weekly meetings.

Developing presentation and communication skills. The graduate students and postdocs will give presentations in the PI research group meetings and seminars. The PI, graduate students, postdocs, and other faculty will provide feedback that will help the students improve the effectiveness of their presentations.

Constant feedback. Feedback is key for guiding graduate students and postdocs. We provide individual feedback to our graduate students and postdocs, assessing their progress and offering guidance on topics that need improvement. Feedback meetings may happen at any time—and can be requested by the student—however, we formally have individual feedback sessions every semester.

Career counseling. From the beginning, career will be a central conversation topic between PIs and students. We will help the student to set their career goals in a variety of areas, and help them build different areas, including publishing papers, networking with peers, or giving effective research and/or classroom presentations.

Helping to build a strong publication record. The PIs will advise the students in planning publications in workshops, conferences, and journals, according to their progress in the program. This will help them develop their research programs and make them known among peers in the field. Through writing papers with the PIs, the student is expected to improve their critical thinking, writing, and scientific skills.

Developing teaching skills. The students will be given the opportunity to guest lecture in classes and to help develop class curricula as well as lecture schedules and student examination methods. They will be offered practice to craft exercises and be involved in grading. Lectures given by the student will be attended by the PIs, who will provide constructive feedback.

The effectiveness of this mentoring plan will be evaluated by monitoring the progress of the students toward the goals and gathering feedback from the students themselves. Adjustments to individual parts of this plan will be made as needed.

Dr. Steinmacher graduated 2 PhDs and co-supervised 5 and was the supervisor of the recipient of the prestigious ACM SIGSOFT Distinguished Dissertation Award 2024 (received by Bianca Trinkenreich, who is a PI in this project). Dr. Trinkenreich recently mentored 80 students from Latin America during the 4th LATAM School in Software Engineering, guiding them in conducting research studies and publishing papers in top-tier venues. This initiative aimed to enhance the representation and participation of underrepresented countries from the Global South in software engineering conferences.

List of Project Personnel and Partner Institutions

1. Igor Steinmacher; Northern Arizona University (NAU); PI
2. Bianca Trinkenreich; Colorado State University (CSU) and The Linux Foundation Research; PI
3. Fabio Marcos de Abreu Santos; Colorado State University (CSU); Co-PI
4. Fabio Calefato; University of Bari; PI - MUR collaborator
5. Filipo Lanubile; University of Bari; MUR collaborator
6. Nicole Novielli; University of Bari; MUR collaborator
7. TBD; Northern Arizona University (NAU); GRA
8. TBD; Colorado State University (CSU); GRA
9. Oliver Kopp; JabRef; Maintainer – Letter of Collaboration
10. Nick Vidal; Open Source Initiative/ClearlyDefined; Community Maintainer – Letter of Collaboration

Suggested Reviewers

1. Preetha Chaterjee, Drexel University, preetha.chaterjee@drexel.edu
2. Sonia Haiduc, Florida State University, shaiduc@fsu.edu
3. Brittany Johnson, George Mason University, bjohnso@gmu.edu
4. Mohamed Mkaouer, University of Michigan, mmkaouer@umich.edu
5. Andy Begel, Carnegie Mellon University, abegel@cmu.edu
6. Tom Zimmermann, University of California, Irvine, tzimmer@uci.edu

Facilities, Equipment and Other Resources

Office, Laboratory, Computing, and Storage: The research will be conducted primarily in a laboratory that includes a workspace for students and a group meeting space. Equipment and furniture include printers, a scanner, ample Ethernet outlets for internet access, and multiple desk areas. All project personnel will have access to a network-accessible file server space and backup services through the School of Informatics, Computing, and Cyber Systems NAU ITS; ICP's Synology file server (12 TB available) will easily satisfy the storage needs of this project. The PI has also access to several laboratories and research facilities available at NAU to perform the proposed research. The school's research laboratories are housed in a 45K sq ft building centrally located on NAU's campus. Built in 2011, SICCS has undergone significant renovations in 2014 and 2016 to support computing-related research, including 10 laboratories, cubicle spaces for graduate students, and meeting and seminar rooms. SICCS also benefits from 100 GB/s networking with NAU's Monsoon computational cluster and storage (see below).

Monsoon: The investigators and students will have access to Monsoon, NAU's capacity-type, Linux-based computer cluster with 808 Intel Xeon cores, 12 TB of memory, and 4 Nvidia Tesla K80 GPU cores. It has been designed to be flexible and handle a diverse set of research requirements. 29 individual systems are interconnected via FDR Infiniband at a rate of 56 Gbps and $<0.07\mu\text{s}$ latency. Cluster nodes have access to 750 TB of shared storage via FDR Infiniband. Monsoon has an estimated peak compute performance of 20 teraflops.

Library Resources: The Cline Library is located on campus and is more than 200,000 square feet and includes a four-hundred-seat assembly hall. The library includes more than 1,400,00 volumes, including 560,000 books; 240,000 government documents; 35,000 maps; 373,000 microforms, 18,000 sound recordings, 10,000 videos, films, and other non-print-media; and 136,000 bound periodical volumes.

Sponsored Projects Management: Northern Arizona University established the Office of Sponsored Projects to aid university faculty with grant submissions and the post-award management of resulting awards. This office has experienced Grants and Contracts Specialists, who help with all grant submissions, including the creation of grant budgets that reflect the needs of the project. The Office also has experienced post-award account managers who ensure that grant account expenditures comply with sponsor rules and regulations. The Office personnel have many years of experience, especially with federal sponsor submissions and post-award management, and can assist researchers with federal grants issues, both pre- and post-award.

Intellectual Resources: Northern Arizona University (NAU) is a high-research institution known for its leadership in environmental sustainability, innovative education, allied health professional training, and Native American educational partnerships. Located in Flagstaff, AZ, NAU serves approximately 15,000 students on its main campus and an additional 10,000 students online or at 30 satellite campuses.

The School of Informatics, Computing, and Cyber Systems (SICCS) offers a premier PhD program in applied interdisciplinary computer sciences. SICCS integrates educational programs and research at the intersection of computational and data sciences with engineering, natural and social sciences, arts, and business, emphasizing environmental informatics, health informatics, computer science, and electrical engineering.

Collaboration with organizations. As evidenced by the letters of collaboration, we will work closely with the Open Source Initiative, Clearly Defined, and JabRef. They will offer support for the project.